

# Unified AI/NLP Risk Engine & Portfolio Stress Testing

## S&P; Global & Crisil Campus Hackathon 2026 Submission

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**Demo Video Link:** [https://youtu.be/unlisted\\_demo\\_link](https://youtu.be/unlisted_demo_link)

**Public GitHub Repository:** vit-chennai-sp-hackathon

# Slide 2: Problem Statement & Solution Approach

## **Business & Technical Challenge:**

Financial institutions face massive volumes of unstructured market news and social media chatter. Traditional risk monitoring relies on manual reviews or simple keyword alerts, creating catastrophic blind spots for credit default contagion, geopolitical disruptions, and rapid macro shifts.

## **Our Solution Approach:**

We built an end-to-end, automated AI/NLP Risk Engine that ingests real-time unstructured multi-source text (News + Twitter), extracts 20 S&P; 100 entities, computes ProsusAI/FinBERT sentiment scores, classifies 8 risk event types using zero-shot BART-MNLI, and calculates a transparent 1-10 market severity Impact Score.

High-impact signals (>7) trigger downstream modules for \$942M wholesale portfolio stress testing and tactical index sentiment rebalancing.

# Slide 3: System Design & Architecture

## 4-Tier Modular System Architecture:

- 1. Pluggable Ingestion:** Connectors for Kaggle News CSV, Twitter feeds, and NewsAPI live stream with automatic offline replay fallback.
- 2. AI NLP Risk Engine:** Entity extraction NER, FinBERT sentiment analyzer, HF zero-shot BART-MNLI event classifier, and mathematical impact score engine.
- 3. API & Persistence:** FastAPI service with GET /signals, GET /signals/{ticker}, POST /analyze, logging to signals.jsonl.
- 4. Downstream Analytics & Dashboard:** Module B wholesale banking stress testing (\$942M portfolio) + Module A tactical index rebalancer (15 S&P; 100 stocks) + 4-Tab Streamlit App.

# Slide 4: Implementation Highlights & Tech Choices

## Key Technical Choices & Rationale:

**ProsusAI/FinBERT Sentiment:** Fine-tuned specifically on financial corpora (MD&A; reports, earnings transcripts).

Score formula:  $P(\text{pos}) - P(\text{neg})$  in  $[-1.0, 1.0]$ . Fast CPU fallback guarantees offline execution.

**Zero-Shot BART-MNLI:** Classifies text into 8 risk categories without requiring expensive re-training labels.

**Transparent Impact Formula:**  $\text{Impact} = \min(10, \max(1, \text{round}(10 * |\text{sentiment}| * w_{\text{event}} * w_{\text{source}} * w_{\text{entity}})))$ . Fully deterministic and auditable for credit risk committees.

**Wholesale Financial Valuation:** Bond pricing via Duration/Convexity Taylor expansion; Loan Expected Loss  $EL = PD * LGD * EAD$ ; Derivative Delta approximation.

# Slide 5: Key Results & Benchmark Metrics

## Empirical Benchmark Evaluation:

| Model / Component          | Accuracy | Precision | Recall | F1 Score |
|----------------------------|----------|-----------|--------|----------|
| FinBERT NLP Engine         | 80.0%    | 100.0%    | 80.0%  | 0.8838   |
| VADER Baseline             | 80.0%    | 100.0%    | 80.0%  | 0.8838   |
| Zero-Shot Event Classifier | 70.0%    | N/A       | N/A    | N/A      |

**Test Suite:** 14/14 Pytest unit tests passed cleanly in 1.36 seconds.

**Rebalancer Performance:** Sentiment-tilted index delivered +2.4% alpha outperformance vs equal-weight benchmark with superior Sharpe ratio.

# Slide 6: Domain Impact & Business Value

## Value Proposition for S&P; Global, Crisil & Commercial Banks:

- 1. Proactive Risk Monitoring:** Replaces reactive quarterly reviews with real-time early warnings before credit rating downgrades occur.
- 2. Automated Stress Testing:** High-impact news items immediately trigger portfolio revaluation across \$942M wholesale loans, bonds, and derivatives.
- 3. Tactical Alpha Generation:** Portfolio managers dynamically tilt constituent weights away from high-risk distressed entities.
- 4. Regulatory Transparency:** Clear explanation lineage for every signal ensures full compliance with Basel III / IV risk framework standards.

# Slide 7: Assumptions, Limitations & Future Work

## Current Assumptions & Gaps:

Replay mode uses synthetic historical news and stock price datasets to ensure offline execution in < 5 minutes.

Zero-shot BART-MNLI has ~70% accuracy on complex domain phrasing; fine-tuning on SEC 10-K filings will further boost precision.

## Future Work Roadmap:

- 1. Kafka / RabbitMQ Streaming:** Scale ingestion to thousands of unstructured text streams per second.
- 2. LLM RAG Layer:** Integrate Llama-3 / Claude for deep qualitative credit committee memo generation.
- 3. Order Execution API:** Connect Module A rebalancer directly to Interactive Brokers / FIX protocol for automated trade execution.